MIL-S-63256C (AR)
AMENDMENT 1
16 February 1990

MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143 (M143E1)
LOGIC CIRCUIT HYBRID FOR

This amendment forms a part of Military Specification MIL-S-63256C (AR), dated 4 February 1987, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, last line: Delete "activitiy" and substitute "activity".

PAGE 3

3.1.4: Delete title and text and substitute "Case and Cover Material and Finish. The case and cover material shall conform to one of the following:"

"3.1.4,a: Delete first sentence and substitute "Iron, Nickel, Cobalt Alloy (KOVAR) per MIL-I-23011, Class I, ASTM F-15."

3.1.5: Delete first sentence and substitute "The lead material shall consist of Iron, Nickel, Cobalt alloy (KOVAR) per MIL-I-23011, Class I, ASTM F-15.".

PAGE 4

3.2.2, line 1 & 2: Delete "When specified in the procurement document" and substitute "Prior to First Article Submission".

PAGE 5

3.3.1, Table I, Pin 16 to 15, under Requirements: Delete "Less than 19 meg" and substitute "Less than 500k".

3.3.1, Table I, Pin 15 to 16, under Requirements: Delete "Greater than 20 meg" and substitute "Greater than 500k".

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PAGE 6

3.3.2.6.1, line 2: Delete "3.5.2.1" and substitute "3.3.2.1".

PAGE 7

3.3.3.1, Table III, d, under Requirements: Delete "Ø.5 to +Ø.5" and substitute "-Ø.5 to +Ø.5".

PAGE 8

3.4.2, line 3: Delete "3" and substitute "2".

3.4.4.1, line 3: Delete "C2" and substitute "C".

3.4.4.2, line 3: Delete "A1" and substitute "A".

PAGE 9

3.4.5, Add to end of paragraph: "The 5 pulses shall be applied in each direction of three principal areas for a total of 3Ø pulses."

3.4.7, line 3: Delete "A" and substitute "B".

PAGE 1Ø

3.6, last line: Delete ""lead integrity"" and substitute "3.4.8".

3.6.1.2, line 4: Delete "i.e." and substitute "e.g.".

PAGE 12

3.9, line 2: Delete "if requested by" and substitute "to".

3.9, last line: Delete "seperate" and substitute "separate".

PAGE 14

4.3.4, Sheet 1 of 2, Delete in its entirety and replace with included sheet.

PAGE 17

4.4.2.1, Sheet 1 of 1, Major 111 and 112, under AQL: Delete "Ø.65%" and substitute "1ØØ%".

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.3.4	Logic Circuit Hybrid				See 4.3.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group I - 1/ 5/	15			
	Physical dimensions			3.	Dwg 9311370/Gage
	Marking permanence			3.5.2	MIL-STD-202
	External visual			3.4.12	MIL-STD-882
	Solderability 2/			3.4.17	MIL-STD-883
	Lead integrity			3.4.8	MIL-STD-883
	Seal			3.4.4	MIL-STD-883
	Bond strength 3/			3.4.9	MIL-STD-883/4.5.1
	Group II - 1/	15			
	Thermal shock			3.4.5	MIL-STD-883
	Temperature cycle			3.4.2	MIL-STD-883
	Seal			3.4.4	MIL-STD-883
	External visual			3.4.12	MIL-STD-883
	Final electrical			3.4.15	4.4.2.2/Test

NOTES: 1/ Each unit in the group shall be subjected to all the examinations or tests in that group unless otherwise specified.
2/ An LTPD-15 (Lot Tolerance Percent Defective) applied to hybrid leads per MIL-M-38510 Table B-1, shall be used for ACC/REJ limits. Minimum sample size shall be 2 hybrids.
3/ An LTPD-7 shall be applied to the wirebonds, with minimum sample size of 2 hybrids. Equal number of wirebonds shall be tested of each, initially.

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PAGE 18

4.4.2.2, Sheet 1 of 1, Major 101, under Paragraph Reference/ Inspection Method: Delete ", Fluke 8000A".

PAGE 19

4.4.2.3, Sheet 1 of 2, Delete in its entirety and replace with included sheet.

PAGE 20

4.4.2.3, Sheet 2 of 2, Delete in its entirety and replace with included sheet.

PAGE 21

4.5.1, line 2: Delete "determine the minimum strength" and substitute "destruction".

4.5.1, last sentence: Delete in its entirety.

PAGE 22

6.2.2, line 3 & 4: Delete "Dover, New Jersey 07801-5000" and substitute "Picatinny Arsenal, New Jersey 07806-5000".

6.3, last line: Delete "Dover, New Jersey 07801-5000" and substitute "Picatinny Arsenal, New Jersey 07806-5000".

PAGE 23

6.5, after M143: Add "(M143E1)".

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63256C (AR)
AMENDMENT 1

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Logic Circuit Hybrid (Lot Acceptance Test)	1 OF 2	9311370
			NEXT HIGHER ASSEMBLY 9311372

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Groups I Through III Inspection Lot Size — Sample Size 1-250 — 2 251-575 — 4				
Major 101	Group I (Non Destructive) 1/ Physical dimensions	100%	3/	3.	Dwg 9311370/Gage
102	Group II (Destructive) 1/ Visual - Internal 2/	100%	3/	3.4.11	MIL-STD-883
103	Group III (Destructive) Solderability - 25% of the terminations	50%	3/	3.4.17	MIL-STD-883
104	Bond strength - 15 bonds shall be pulled to destruction for information on each unit.	100%	3/	3.4.9	MIL-STD-883/4.5.1

NOTES:
1/ Group I samples may be used in Group IV or V. Electrical rejects may be used for Group II.
2/ Requires removal of Cover (Hybrids may be cleaned).
3/ Accept on 0 defects, reject on 1 or more defects.

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5 (16)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Logic Circuit Hybrid (Lot Acceptance Test)	2 OF 2	9311370
			NEXT HIGHER ASSEMBLY: 9311372

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major	Groups IV 2/				
	Inspection Lot Size: Sample Size				
	1-100: 3				
	101-250: 5				
	251-475: 7				
	Group IV (Destructive) 1/				
105	Thermal shock	100%	3/	3.4.5	MIL-STD-883
106	Temperature cycling	100%	3/	3.4.2	MIL-STD-883
107	Seal	100%	3/	3.4.4	MIL-STd-883
108	Electrical tests	100%	3/	3.4.15	Test/4.4.2.2
	Group V - 2/				
	Inspection Lot Size: Sample Size				
	1-100: 3				
	101-250: 5				
	251-475: 7				
	Group V (Destructive) 1/				
109	Mechanical shock	100%	3/	3.4.6	MIL-STD-883
110	Vibration variable frequency	100%	3/	3.4.7	MIL-STD-883
111	Seal	100%	3/	3.4.4	MIL-STD-883
112	Electrical tests	100%	3/	3.4.15	Test/4.4.2.2

NOTES:
1/ Each unit in group shall be subjected to all the examinations or tests in that group.
2/ Groups IV & V require separate samples, group samples shall not be intermixed.
3/ Accept on 0 defects, reject on 1 or more defects.

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